

STA201 Assignment 2 Solution (Spring 2023)

Question 1

Blood cocaine concentration (mg/L) was determined both for a sample of individuals who had died from cocaine-induced excited delirium (ED) and for a sample of those who had died from a cocaine overdose without excited delirium; survival time for people in both groups was at most 6 hours. [9]

ED:

0	0	0	0	0.1	0.1	0.1	0.1	0.2	0.2
0.3	0.3	0.3	0.3	0.4	0.4	0.5	0.5	0.5	0.5
0.6	0.7	0.7	0.8	0.9	1	1.5	2	2.1	2.5
3	3.9	5	8.5	11	12.5	14			

NON-ED:

0	0	0	0.1	0.1	0.2	0.3	0.3	0.3	0.5
0.6	0.6	0.8	0.8	0.8	0.8	1	1.5	2	2.2
2.6	3.3	5	5.5	7	7.7	9	11	12	15
18	19.5	22	24	25.5	25.5	25.5	27	27.5	28

- a) Determine the three-quartile values for blood cocaine concentration for both ED and Non-ED samples.

Answer:

For ED samples, n=37

First quartile, $Q_1 = \frac{1}{4} * 37 = 9.25$ (10th value) = 0.2

Second quartile, $Q_2 = \frac{2}{4} * 37 = 18.5$ (19th value) = 0.5

Third quartile, $Q_3 = \frac{3}{4} * 37 = 27.75$ (28th value) = 2

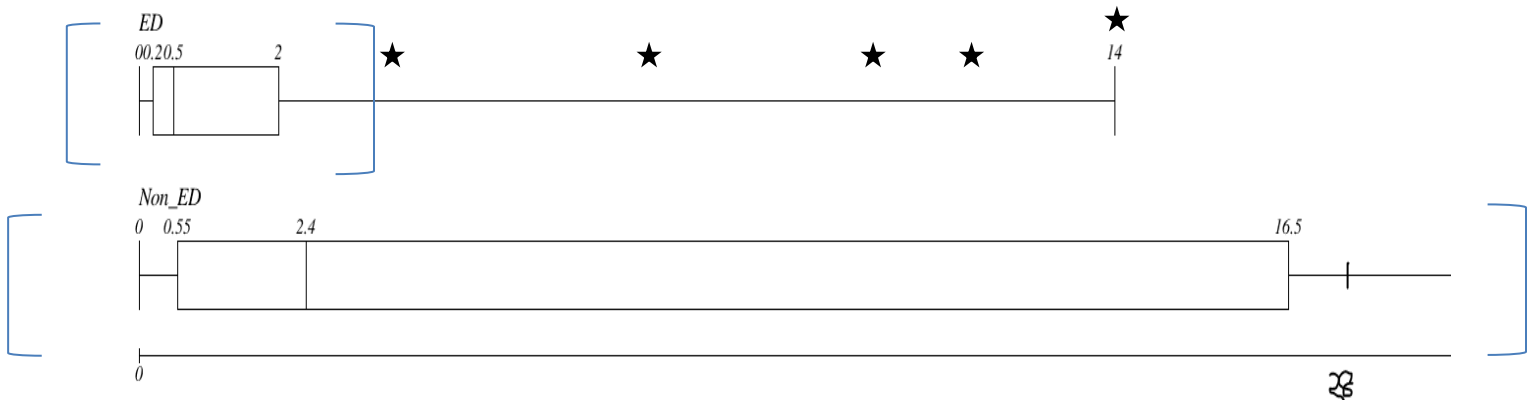
For non-ED samples, n=40

First quartile, $Q_1 = \frac{1}{4} * 40 = 10$ (Average of 10th and 11th observations) = 0.55

Second quartile, $Q_2 = \frac{2}{4} * 40 = 20$ (Average of 20th and 21th observations) = 2.4

Third quartile, $Q_3 = \frac{3}{4} * 40 = 30$ (Average of 30th and 31th observations) = 16.5

b) Construct a comparative boxplot (two boxplots on the same set of axes, one above the other)



c) Describe (Compare and Contrast) the ED and Non-ED samples in your own words from the boxplot.

Comparison between the boxplots:

- Range of non-ED samples greater than ED samples
- The interquartile range of ED samples is less than the interquartile range of non-ED samples
- The middle 50% of the observations of ED samples are more concentrated towards the median than that of the non-ED samples since ED samples have a lower interquartile range than non-ED samples.
- Both box plots show a positively skewed distribution as $Q_3 - Q_2 > Q_2 - Q_1$. However, non-ED samples are more positively skewed.

d) Are any outliers within the data set? If so, include them in the boxplot and identify them. If not, please explain why.

For ED samples,

$$IQR = Q_3 - Q_1 = 2 - 0.2 = 1.8$$

$$\text{Lower Fence (L.F.)} = Q_1 - (1.5 * IQR) = 0.2 - (1.5 * 1.8) = 0.2 - 2.7 = -2.5$$

$$\text{Upper Fence (U.F.)} = Q_3 + (1.5 * IQR) = 2 + (1.5 * 1.8) = 2 + 2.7 = 4.7$$

Here, no values lie outside the lower fence but 5, 8.5, 11, 12.5 and 14 lies outside the upper fence. So, 5, 8.5, 11, 12.5 and 14 are outliers for the ED dataset.

For non-ED samples,

$$IQR = Q_3 - Q_1 = 16.5 - 0.55 = 15.95$$

$$\text{Lower Fence (L.F.)} = Q_1 - (1.5 * IQR) = 0.55 - (1.5 * 15.95) = 0.55 - 23.925 = -23.375$$

$$\text{Upper Fence (U.F.)} = Q_3 + (1.5 * IQR) = 16.5 + (1.5 * 15.95) = 16.5 + 23.925 = 40.425$$

Here, no values lie outside the lower fence -23.375 and outside the upper fence 40.425. So, there are no outliers for the non-ED dataset.

Question 2

A Benchmark company is testing Computer Power Supply Unit (PSU) of two different companies X and Y. PSU's were tested at the 120% stress to check for the overvoltage protection. The following data given in Table, represent time to failure (in hours) for random samples taken from the days' production of the companies. [12.5]

Table: Failure time (in hours) of PSU of X and Y Company

Company	Failure time (in hours)									
X	0.54	1.80	4.52	2.05	1.03	1.18	0.80	1.33	1.29	1.11
	3.34	1.54	0.08	4.12	0.60	0.72	0.92	1.05	1.66	3.03
	1.81	2.17	0.63	0.56	5.03	0.09	0.18	0.34	1.51	1.45
Y	1.52	0.19	1.55	0.02	0.07	3.65	0.40	0.24	1.66	1.45
	1.60	1.80	4.69	0.08	7.89	1.58	0.64	0.03	0.23	0.72

- a) What are the sample size of the company X & Y? [0.5]
- b) Compute the sample mean, variance, and standard deviation for the time to failure for company X. [1+2.5+0.5]
- c) Compute the sample mean, variance, and standard deviation for the time to failure for company Y. [1+2.5+0.5]
- d) Compute the coefficient of variation (CV) for the time to failure for company X. [0.5]
- e) Compute the coefficient of variation (CV) for the time to failure for company Y. [0.5]
- f) Which measure one should consider to compare the performance / consistency of the product of the two company? And why? [2]
- g) Which company's product is more consistent in their performance? [1]

Answer:

- a) Sample size for company X=30
Sample size for company Y=20

b) For company **X**:

$$\text{Sample mean} = \frac{0.54+1.8+4.52+\dots+1.51+1.45}{30} = \frac{46.48}{30} = \mathbf{1.54}$$

$$\text{Variance} = \frac{\sum_{i=0}^n (xi - \bar{x})^2}{n-1} = \frac{47.71}{29} = \mathbf{1.64}$$

$$\text{Standard Deviation} = \sqrt{\text{Variance}} = \mathbf{1.28}$$

c) For company **Y**:

$$\text{Sample mean} = \frac{1.52+0.19+1.55+\dots+0.23+0.72}{30} = \frac{30.01}{20} = \mathbf{1.50}$$

$$\text{Variance} = \frac{\sum_{i=0}^n (xi - \bar{x})^2}{n-1} = \frac{71.65}{19} = \mathbf{3.77}$$

$$\text{Standard Deviation} = \sqrt{\text{Variance}} = \mathbf{1.94}$$

d) Coefficient of Variation, $CV_x = \frac{SD}{\bar{x}} \times 100 = \frac{1.28}{1.54} \times 100 = \mathbf{83.12\%}$

e) Coefficient of Variation, $CV_y = \frac{SD}{\bar{x}} \times 100 = \frac{1.94}{1.50} \times 100 = \mathbf{129.33\%}$

f) The coefficient of Variation (CV)/ Standard Deviation one should consider to compare the performance / consistency of the product of the two company based on the situation.

The **coefficient of variation** represents the ratio of the standard deviation to the mean, and it is a useful statistic for **comparing** the degree of **variation** from one data series to another, even if the means are drastically different from one another.

g) As $CV_x < CV_y$, Company X is more consistent in the performance comparing to company Y.

Question 3

You are given the following data pertaining to kilowatt hours of electricity consumed by some users in the city of Dhaka: [5.5]

Consumption (in Kilowatt hours)	Number of users
20 – 30	13
30 – 40	35
40 – 50	25
50 – 60	23
60 – 70	0 + X

Where **X** is the sum of the **last two digits** of your University ID. For example: if your university ID is: 181234**56** then **X** would be **5 + 6 = 11**. Estimate the following:

- a) sample variance and the sample standard deviation [2.5]
- b) mean deviation [2.0]
- c) coefficient of variation. [1.0]

Answer:

a) Standard deviation and variance

For x = 00

Consumption (in Kilowatt hours)	Mid value	Frequency	$f_i x_i$	$(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^2$	$f_i x_i - \bar{x} $
20 – 30	25	13	325	257.33	3345.35	208.54
30 – 40	35	35	1225	36.50	1277.56	211.45
40 – 50	45	25	1125	15.66	391.71	98.95
50 – 60	55	23	1265	194.83	4481.20	321.04
60 – 70	65	0	0	574.00	0	0
		Total: 96	Total: 3940		Total: 9495.83	Total: 840

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{3940}{96} = \mathbf{41.04}$$

$$\text{Variance} = \frac{\sum fi(xi-\bar{x})^2}{fi-1} = \frac{9495.83}{96-1} = \mathbf{99.95}$$

$$\text{Standard deviation (SD)} = \sqrt{99.95} = \mathbf{9.99}$$

b) $\text{Mean deviation} = \frac{\sum fi|xi-\bar{x}|}{\sum fi} = \frac{840}{96} = \mathbf{8.75}$

c) Coefficient of variation:

$$\text{CV} = \frac{9.99}{41.04} \times 100 = \mathbf{24.36\%}$$

SUM	Mean	Std dev	Var	Mean dev	CV
0	41.04	9.99	99.95	8.75	24.36%
1	41.28	10.23	104.83	8.90	24.79%
2	41.53	10.46	109.48	9.05	25.19%
3	41.76	10.67	113.93	9.18	25.55%
4	42	10.87	118.18	9.32	25.88%
5	42.22	11.05	122.23	9.44	26.18%
6	42.45	11.22	126.11	9.56	26.45%
7	42.66	11.39	129.81	9.67	26.70%
8	42.88	11.54	133.34	9.77	26.92%
9	43.09	11.69	136.72	9.87	27.13%
10	43.30	11.82	139.94	9.97	27.31%
11	43.50	11.95	143.02	10.06	27.48%
12	43.70	12.08	145.96	10.14	27.64%
13	43.89	12.19	148.77	10.22	27.78
14	44.09	12.30	151.45	10.29	27.91
15	44.27	12.41	154.02	10.36	28.02
16	44.46	12.51	156.47	10.43	28.13
17	44.67	12.60	158.80	10.50	28.23
18	44.82	12.69	161.03	10.55	28.31

Question 4

For a distribution Karl Pearson's coefficient of skewness is 0.64, Variance is 40 and mean is $\sqrt{(X + Y + Z)}$ Find mode and median. [Here X,Y,Z = First 3 digits of you Student ID] [3]

Answer:

Suppose the sum of the first 3 digits of your student ID is 6.

$$\text{Mean}, \sqrt{6} = 2.4494$$

$$\text{SD} = \sqrt{40} = 6.324$$

$$\text{Skp} = 0.64$$

Now,

$$\frac{\text{Mean} - \text{Mode}}{\text{SD}} = 0.64$$

$$\text{Mode} = -1.59796$$

And,

$$\frac{3(\text{Mean} - \text{Median})}{\text{SD}} = 0.64$$

$$\text{Median} = 1.1003$$